8A Pythagoras' Theorem

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Find the exact length of the hypotenuse c in each right-angled triangle with shorter sides a and b:

(a) $a = 6\ \text{cm}, b = 8\ \text{cm}$ (b) $a = 5\ \text{m}, b = 12\ \text{m}$ (c) $a = 15\ \text{cm}, b = 20\ \text{cm}$ (d) $a = 9\ \text{m}, b = 40\ \text{m}$

Q2. Find the unknown shorter side a in each right-angled triangle with hypotenuse c and shorter side b:

(a) $c = 10\ \text{cm}, b = 8\ \text{cm}$ (b) $c = 25\ \text{m}, b = 7\ \text{m}$ (c) $c = 34\ \text{cm}, b = 30\ \text{cm}$ (d) $c = 29\ \text{m}, b = 21\ \text{m}$

Q3. Find the unknown side as an exact surd AND rounded to 2 decimal places:

(a) Hypotenuse c when $a = 4\ \text{cm}, b = 7\ \text{cm}$ (b) Hypotenuse c when $a = 6\ \text{m}, b = 9\ \text{m}$ (c) Shorter side a when $c = 11\ \text{cm}, b = 6\ \text{cm}$ (d) Shorter side a when $c = 14\ \text{m}, b = 9\ \text{m}$

Q4. Test whether each set of three side lengths forms a right-angled triangle (Yes or No):

(a) $9\ \text{cm}, 12\ \text{cm}, 15\ \text{cm}$ (b) $7\ \text{cm}, 10\ \text{cm}, 12\ \text{cm}$ (c) $12\ \text{m}, 16\ \text{m}, 20\ \text{m}$ (d) $1.5\ \text{m}, 2.0\ \text{m}, 2.5\ \text{m}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A rectangular soccer pitch at Darebin International Sports Centre measures $105\ \text{m}$ long by $68\ \text{m}$ wide. Find the exact surd length and decimal length (2 d.p.) of a diagonal run from one corner flag to the opposite corner flag.

Q6. A $5.0\ \text{m}$ painter's ladder leans against a vertical house wall with its foot $1.4\ \text{m}$ out from the base of the wall on level ground. How high up the wall does the top of the ladder reach?

Q7. A yacht sails $16\ \text{km}$ due north from Queenscliff into Port Phillip Bay, then turns and sails $12\ \text{km}$ due west. How far is the yacht in a direct straight line from Queenscliff?

Q8. An equilateral triangle has side lengths of $10\ \text{cm}$. By drawing its perpendicular altitude h (which bisects the base into two $5\ \text{cm}$ segments), find the exact surd height h and the height rounded to 2 decimal places.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A square has a diagonal of length $12\ \text{cm}$. Let the side length of the square be s. Using Pythagoras' theorem $s^2 + s^2 = 12^2$, find the exact value of s^2 and explain why s^2 immediately gives you the area of the square without needing to approximate s.

Q10. Show that if (a, b, c) is any Pythagorean triad (so $a^2 + b^2 = c^2$), then (ka, kb, kc) is also a Pythagorean triad for any positive integer k.